Public Economics (ECON 131)
Section #6: Labor Income Responses to Taxation

GSI: Steve Ramos

March 3, 2025

Contents

Just as last section was largely a review of Section 2, but with new interpretations of utility functions and budget constraints, this section is largely a review of Section 5 but with new graphs. We will also distinguish between so-called **real responses**, which change what is actually going on in the economy and **paper responses** which involve the re-labeling or short-term re-timing of realizing income that are more accounting tricks than actual changes in how the economy works.

1 Real Labor Supply Responses

1.1 Theoretical Review

- We saw last section that real labor supply responses are the result of a combination of substitution and income effects
  - Substitution effects are driven by the slope of the budget constraint, inducing people to work more when their hourly wage is higher
  - Income effects are driven by the level of the budget constraint, inducing people to work less when their income is higher
- Responses can occur on both the intensive margin (how many hours) and the extensive margin (whether someone works at all)

1.1.1 Practice Problem

Let's revisit the theory with one another practice problem to tie concepts together:

Assume that individuals have the same utility function over consumption and labor given by:

$$U(c, l) = c + \theta \ln(10 - l)$$

→ constant

where c represents consumption and l represents hours of labor and θ is a given positive parameter. Here, $\ln(x)$ denotes the natural logarithm of x (this can also be denoted by $\log(x)$).Assume also that the only income that individuals have is from labor income and that the hourly wage rate is given by w.

Note: In Section 5, we used l to denote leisure. Here l denotes labor. It useful to know how to do these problems both ways. It is unfortunate, from a notation standpoint, that both labor and leisure begin with l.

1. Write the budget constraint faced by the individual.

   $c = w \cdot l$ → labor

   aggregate consumption; wage

   $P_c = 1$ for simplicity!

2. Set up the maximization problem of this individual and solve for the optimal choices of labor and consumption.

$$\mathcal{L} = \max_{c,l,\lambda} U(c, l) + \lambda[c - wl]$$
$$= \max_{c,l,\lambda} c + \theta \ln(10 - l) + \lambda[c - wl]$$

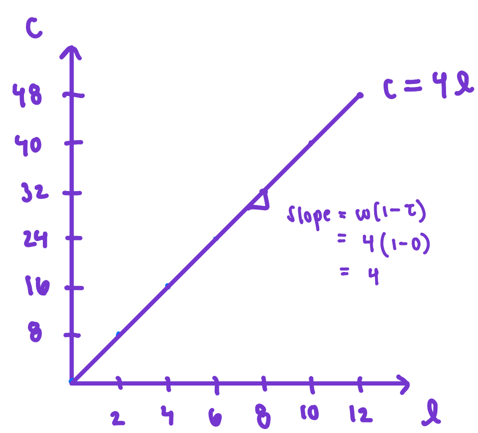
- $\frac{\partial \mathcal{L}}{\partial c} = 1 + \lambda = 0 \Rightarrow \lambda = -1$
- $\frac{\partial \mathcal{L}}{\partial l} = \theta \cdot \frac{1}{10 - l}(-1) + \lambda(-w) = 0$

$$\Rightarrow \frac{\theta}{10 - l} = w \qquad \Rightarrow \frac{\theta}{w} = 10 - l$$
$$\Rightarrow w(10 - l) = \theta \qquad \Rightarrow l^* = 10 - \frac{\theta}{w}$$

- $c = w\left[10 - \frac{\theta}{w}\right]$

  $c^* = 10w - \theta$

3. Suppose now that the market wage w is equal to 4. Draw the consumption-labor trade-off for a representative individual.



if $l = 0$, $c = 0$

if $l = 12$, $c = 48$

Suppose that the government introduces an Earned Income Tax Credit such that for the first \$24 in earnings, the government pays 50 cents per dollar on wages earned. After \$24 the credit is reduced at a rate of 50 cents per dollar earned. When the credit reaches zero, there is no additional EITC.

4. Draw the relationship between labor and consumption, and between pre-tax/transfer income and post-tax/transfer income.

Phase In

when does someone earn \$24

$24 = 4\ell$

$\ell = 6$

after-transfer wage $\tau = -0.5$

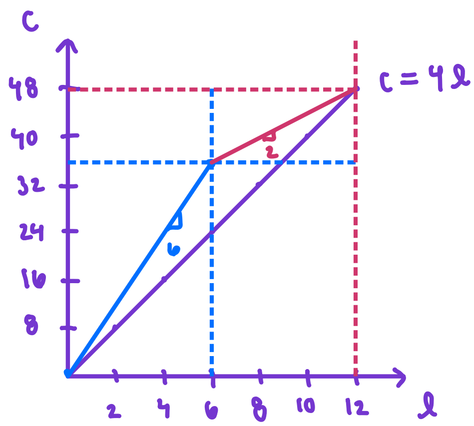
$c = w(1 - \tau)\ell$

$= 4(1 + 0.5)\cdot 6$

$= 6 \cdot 6$

$= 36$

slope = 6



relationship b/w labor + consumption

Phase Out "claw backs"

$\tau = 0.5$

linear equations

$4\ell = 36 + 4(1 - 0.5)[\ell - 6]$

$4\ell = 36 + 2[\ell - 6]$

$4\ell = 36 + 2\ell - 12$

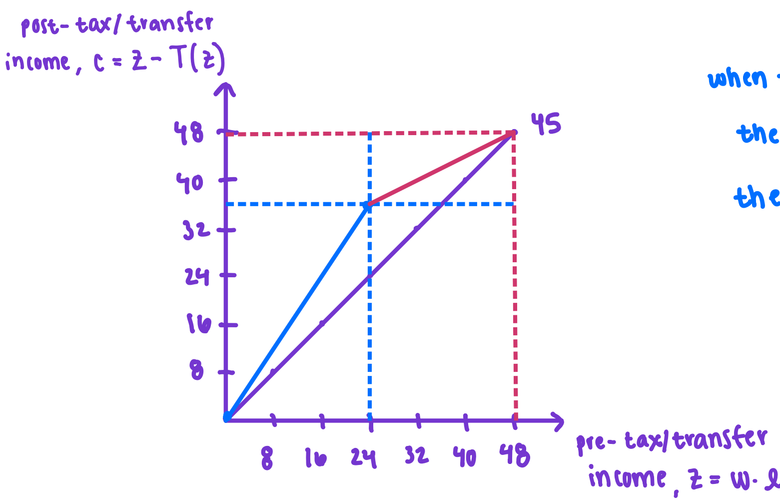
$2\ell = 24$

$\ell = 12$

$c = 4(12) = 48$

Slope = 2

relationship b/w pre-tax/transfer income and post-tax/transfer income



when they work six hours,

they earn 24 w/ no transfer

they earn 36 w/ transfer

$= 24 + 4(0.5)(6)$

$= 24 + 2(6)$

$= 24 + 12$

5. Consider now two individuals with different relative valuations of labor and consumption. Anna has a θ of 12 and Bob has a θ of 4. For both these individuals find the optimal choice of labor in the no-tax context and under the EITC setting previously introduced.

Recall our optimization: $c^* = 10w - \theta$, $\ell^* = 10 - \frac{\theta}{w}$

w/ no tax $w = 4$

Anna ($\theta = 12$)

$\ell^* = 10 - \frac{12}{4} = 10 - 3 = 7$ $\quad c^* = 10(4) - 12 = 40 - 12 = 28$

Bob ($\theta = 4$)

$\ell^* = 10 - \frac{4}{4} = 10 - 1 = 9$ $\quad c^* = 10(4) - 4 = 40 - 4 = 36$

Under EITC

→ **Phase In** $w = 6$

Anna ($\theta = 12$)

$\ell^* = 10 - \frac{12}{6} = 10 - 2 = 8$ $\quad c^* = 10(6) - 12 = 60 - 12 = 48$

Bob ($\theta = 4$)

$\ell^* = 10 - \frac{4}{6} = \frac{60}{6} - \frac{4}{6} = \frac{56}{6} = 9.33$ $\quad c^* = 10(6) - 4 = 60 - 4 = 56$

→ **Phase Out** $w = 2$

Anna ($\theta = 12$)

$\ell^* = 10 - \frac{12}{2} = 10 - 6 = 4$ $\quad c^* = 10(2) - 12 = 20 - 12 = 8$

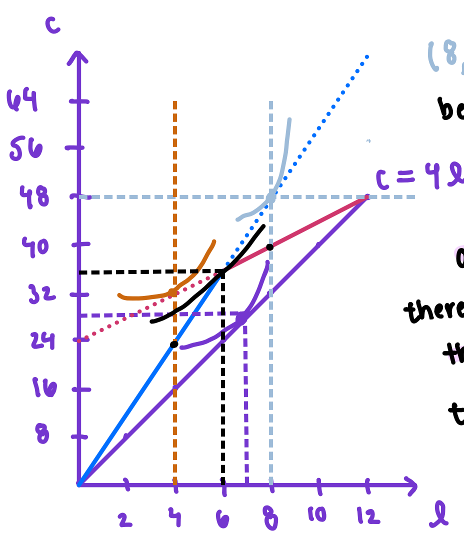
Bob ($\theta = 4$)

$\ell^* = 10 - \frac{4}{2} = 10 - 2 = 8$ $\quad c^* = 10(2) - 4 = 20 - 4 = 16$

when $w = 2$, we get small numbers, this would be fine if the y-intercept of the phase out period was 0 [$c = 0$, $\ell = 0$], but that's not the case here! $y = mx + b$

$36 = 2(6) + b \Rightarrow b = 36 - 12 = 24$

$c^* = 8 + 24 = 32$ $\quad c^* = 16 + 24 = 40$



Anna ($\theta = 12$)

↳ reduces their labor supply from 7 hours to 6 hours

(8, 48) & (4, 32) would be the optimal, but NOT on the budget constraint; Lagrangian only finds optimal interior points! therefore, Anna bunches @ the kink point! (6, 36)

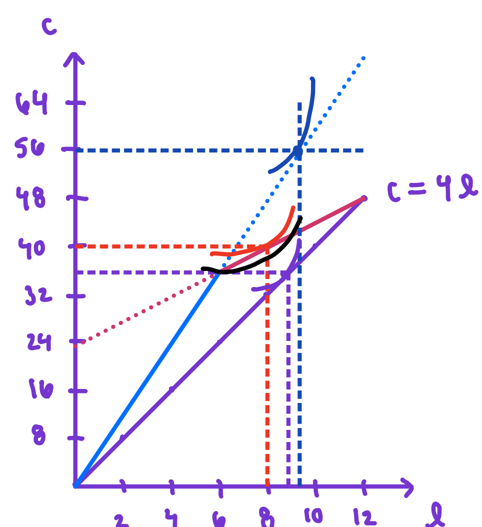
they could pick other points like (8, 40) & (4, 24) but $U(36,6) > U(40,8) > U(24,4)$

first, (9.33, 56) is NOT on the budget constraint! however, (8, 40) is, so that is the optimal solution! the kink point is not optimal here, which you can verify by comparing the utilities. $U(40, 8) > U(36, 6)$

$40 + 4\ln(10-8) > 36 + 4\ln(10-6)$

$40 + 4\ln(2) > 36 + 4\ln(4)$

$42.773 > 41.545$



Bob ($\theta = 4$)

↳ reduces their labor supply from 9 hours to 8 hours

6. Plot the solutions you found in part (f) in the graph you draw in part (e).

7. Note that both Anna and Bob reduce their labor supply in response to the tax. If you would like an example of someone who increases her labor supply after the EITC, consider Ciara with $\theta = 20$. What are is her optimal labor supply with or without the EITC?

w/ no tax $w = 4$

$\ell^* = 10 - \frac{20}{4}$
$= 10 - 5$
$= 5$

$c^* = 10(4) - 20$
$= 40 - 20$
$= 20$

w/ tax $w = 6$

$\ell^* = 10 - \frac{20}{6}$
$= \frac{60}{6} - \frac{20}{6}$
$= 6.66$

$c^* = 10(6) - 20$
$= 60 - 20$
$= 40$

NOT on the budget constraint!

w/ tax $w = 2$

$\ell^* = 10 - \frac{20}{2}$
$= 10 - 10$
$= 0$

$c^* = 10(2) - 20$
$= 0$

$c^* = 24 + 2(0)$
$= 24$

also NOT on the budget constraint!

therefore the kink is optimal @ (6, 36)

before 5 hours, now 6 hours, so Ciara ↑ labor supply!

1.2 Empirical Evidence

- The Congressional Budget Office uses a population-wide labor supply elasticity of 0.1-0.3. An elasticity of 0.1 implies that doubling wages would increase hours worked by 10%.
- The substitution effect is larger than the income effect, but both are significant.
- Historically, earnings the elasticity for married women was larger than for married men, but they are converging and currently fairly close.
- **High income**: Top 1% income share seems to respond to top tax rate in the US, but this correlation is largely absent at lower incomes (eg. next 9%). Looking across countries, there is a strong negative correlation between top income shares and top tax rates. There is, however, mixed evidence on their labor supply elasticites, mostly pointing toward high-earners having similar elasticities to the rest of the income distribution.
- **Low income**: There is evidence of a strong extensive margin response to the EITC but not an intensive margin response (except for self-employed individuals)
- Card & Hyslop (2005) study the effect of a randomly assigned welfare earnings subsidy in Canada. They find there is a strong employment effect in response to the subsidy, and the employment effect vanishes once the subsidy does.
- Kleven (2018) studies the effect of the EITC relative to traditional welfare by comparing labor force participation of single women with and without children. He finds a dramatic increase in labor force participation of single mothers (most likely driven by changing social norms and economic upturn more than EITC and welfare reform).

2 Evasion and Avoidance Responses

2.1 Theory

- Individuals can shift income over time (eg from this year to next) or over tax bases (from corporate to individual or vice versa, or from one legal jurisdiction to another) .

2.2 Empirical Evidence

- Saez (2010): Bunching at the first kink of the EITC seems to be primarily driven by self-employed individuals, and a randomly chosen individual is more likely to report earnings that maximize their EITC refund if they live in a zip code with sharp bunching of self-employed individuals, suggesting that information plays a role in people's ability to evade taxes.
- Goolsbee (2000): executives re-time their stock option exercises in response to a 1993 change in top marginal income tax rates.
- Kleven et al (2013a), Kleven et al (2013b), Moretti (2017): high earners seem to have a high elasticity of migration with respect to the net-of-tax rate.